
Snowflake ARRAY Functions – Complete SQL Examples (with Output)

◆ ARRAY CREATION

```
-- ARRAY_CONSTRUCT
SELECT ARRAY_CONSTRUCT(1, 2, 3);
-- Output: [1,2,3]

-- ARRAY_CONSTRUCT_COMPACT (removes NULLs)
SELECT ARRAY_CONSTRUCT_COMPACT(1, NULL, 3);
-- Output: [1,3]
```

◆ AGGREGATION

```
-- ARRAY_AGG
SELECT ARRAY_AGG(id) FROM VALUES (1),(2),(3) t(id);
-- Output: [1,2,3]

-- ARRAY_UNION_AGG
SELECT ARRAY_UNION_AGG(arr)
FROM VALUES ([1,2]),([2,3]) t(arr);
-- Output: [1,2,3]

-- ARRAY_UNIQUE_AGG
SELECT ARRAY_UNIQUE_AGG(id)
FROM VALUES (1),(2),(2),(3) t(id);
-- Output: [1,2,3]
```

◆ ADD / MODIFY ELEMENTS

```
-- ARRAY_APPEND
SELECT ARRAY_APPEND([1,2], 3);
-- Output: [1,2,3]

-- ARRAY_PREPEND
SELECT ARRAY_PREPEND(0, [1,2]);
-- Output: [0,1,2]

-- ARRAY_INSERT
SELECT ARRAY_INSERT([1,3], 1, 2);
-- Output: [1,2,3]
```

```
-- ARRAY_REMOVE
SELECT ARRAY_REMOVE([1,2,3], 2);
-- Output: [1,3]

-- ARRAY_REMOVE_AT
SELECT ARRAY_REMOVE_AT([1,2,3], 1);
-- Output: [1,3]

-- ARRAY_REPEAT
SELECT ARRAY_REPEAT('A', 3);
-- Output: ['A','A','A']
```

◆ COMBINE & TRANSFORM

```
-- ARRAY_CAT
SELECT ARRAY_CAT([1,2], [3,4]);
-- Output: [1,2,3,4]

-- ARRAY_FLATTEN
SELECT ARRAY_FLATTEN([[1,2],[3,4]]);
-- Output: [1,2,3,4]

-- ARRAY_SLICE
SELECT ARRAY_SLICE([1,2,3,4], 1, 3);
-- Output: [2,3]

-- ARRAY_REVERSE
SELECT ARRAY_REVERSE([1,2,3]);
-- Output: [3,2,1]

-- ARRAY_SORT
SELECT ARRAY_SORT([3,1,2]);
-- Output: [1,2,3]

-- ARRAY_TO_STRING
SELECT ARRAY_TO_STRING([1,2,3], '');
-- Output: '1,2,3'
```

◆ FILTERING & CLEANING

```
-- ARRAY_COMPACT
SELECT ARRAY_COMPACT([1,NULL,2]);
-- Output: [1,2]

-- ARRAY_DISTINCT
SELECT ARRAY_DISTINCT([1,2,2,3]);
-- Output: [1,2,3]
```

◆ SEARCH & ANALYSIS

```
-- ARRAY_CONTAINS
SELECT ARRAY_CONTAINS(2, [1,2,3]);
-- Output: TRUE

-- ARRAY_POSITION
SELECT ARRAY_POSITION(2, [1,2,3]);
-- Output: 1

-- ARRAY_SIZE
SELECT ARRAY_SIZE([1,2,3]);
-- Output: 3

-- ARRAY_MAX
SELECT ARRAY_MAX([1,5,3]);
-- Output: 5

-- ARRAY_MIN
SELECT ARRAY_MIN([1,5,3]);
-- Output: 1
```

◆ SET OPERATIONS

```
-- ARRAY_INTERSECTION
SELECT ARRAY_INTERSECTION([1,2,3], [2,3,4]);
-- Output: [2,3]

-- ARRAY_EXCEPT
SELECT ARRAY_EXCEPT([1,2,3], [2]);
-- Output: [1,3]

-- ARRAYS_OVERLAP
SELECT ARRAYS_OVERLAP([1,2], [2,3]);
-- Output: TRUE
```

◆ ADVANCED / UTILITIES

```
-- ARRAY_GENERATE_RANGE
SELECT ARRAY_GENERATE_RANGE(1, 5);
-- Output: [1,2,3,4]

-- ARRAYS_TO_OBJECT
SELECT ARRAYS_TO_OBJECT(['a','b'], [1,2]);
-- Output: {"a":1,"b":2}

-- ARRAYS_ZIP
SELECT ARRAYS_ZIP([1,2], ['a','b']);
-- Output: [{"$1":1,"$2":"a"}, {"$1":2,"$2":"b"}]
```

```
SELECT VALUE  
FROM TABLE(FLATTEN(INPUT => ARRAY_CONSTRUCT(1,2,3)));  
-- Output:  
-- 1  
-- 2  
-- 3
```

